



مستشفى رويال حياة
ROYALE HAYAT HOSPITAL

The Power of Pink

LOVE,
HOPE,
FAITH

Breast Cancer Awareness Handbook



Table of Content

Breast Cancer Awareness Handbook – “The Power of Pink Journey”

1. Family Clinic
2. Breast Cancer Test
3. Surgery Department – Oncology
 - a. Breast Clinic Guide for Patients
 - b. Breast Self-Examination
 - c. Screening for Breast Cancer
 - d. Surgical Options in the Treatment of Breast Cancer
 - e. Post Breast & Axillary Surgeries
 - f. Breast Biopsies
 - g. Chemotherapy
 - h. Radiation Therapy
 - i. Hormonal Therapy
4. Diagnostic Imaging
5. Nutrition Care
6. Royale Hayat Dental
7. La Cosmetique Royale
8. Royale Home Health Breast Cancer Program





Family Clinic

1

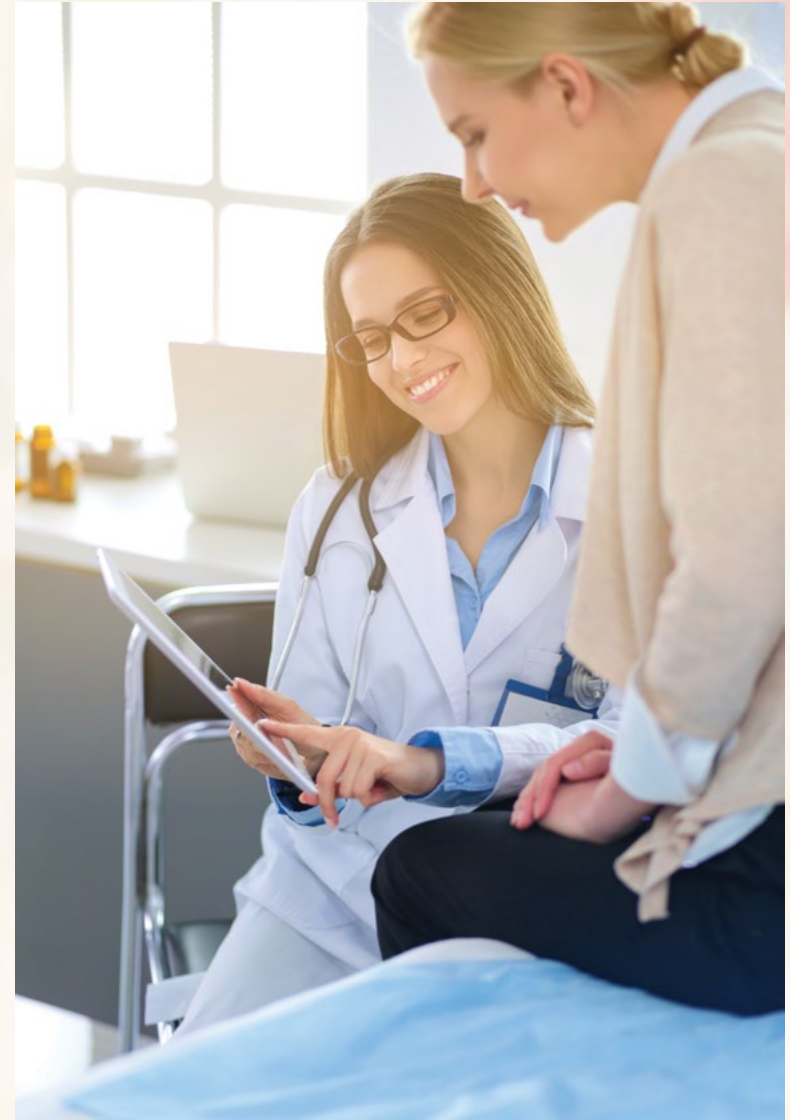
Family Clinic

Upon arrival at the family clinic, the patient is initially greeted by our reception staff, who communicate with the clinic nurse to inform them of the patient's arrival. The clinic nurse invites the patient to perform the basic vitals check. This includes blood pressure, pulse, temperature, oxygen saturations, weight, height and BMI. The patients are now ready to be seen by the doctor.

An initial consultation with the family doctor involves a detailed assessment of the patient's background and past medical history, as well as their presenting complaint, family history, medication, allergies and social history.

Following history and assessment, the doctor will proceed to a full general examination including the heart, lungs, abdomen/pelvis, breasts and focus on the areas of patient's presenting complaint.

The family doctors focus on the patient as a whole and are able to assess multiple co-morbidities. Patient education and screening is paramount in family medicine and we encourage and teach our patients self-examinations including self-breast examinations. The consultation then culminates in a mutually agreed management plan and follow up arrangements.





Breast Cancer Tests

2



Breast cancer tests

Can be done from simple tests to biochemical tumor markers and detection of high-risk patients for breast cancer can be done by genetic testing for those with extensive family history of breast cancer. Screening is done with mammographic imaging and histopathology testing if the tumor is present. Tumor markers are unfortunately not reliable in diagnosing breast cancer. The following tests can be done for breast cancer:

- CBC.
- CA15.3.
- CEA.
- Breast Cancer Screen (BRCA1 and BRCA2).
- FNAC.
- Histopathology Testing.

CBC: A complete blood count (CBC) can uncover anemia which can be seen in malignancy.

CEA and CA 15.3: are tumor markers that can be elevated in breast cancer.

Genetic screening: is most helpful in familial cancers

Sequencing of the 2 genes BRCA 1 and BRCA 2 is done to determine risk of developing breast cancer. Mutations in these genes indicate an elevated risk of malignancy in the individual. It also necessitates screening of all family members.

- **FNAC:** to help in differential diagnosis. During a fine needle aspiration (FNA), a small amount of breast tissue or fluid is removed from a suspicious area with a thin, hollow needle to check for cancer cells. This type of biopsy is sometimes an option if other tests raise the suspicion of breast cancer (although a core needle biopsy is often preferred).

- **Pathology testing:** We are using the most advanced technology as recommended by College of American Pathologists (CAP) to help in prognosis (ER, PR and HER-2)

The previously mentioned modalities are present in our lab for diagnosis of breast cancer. In addition, our testing provides valuable information for prognostic and staging and helps the medical team in treatment. The team involved in diagnosing the blood tests and tissue biopsies are board certified laboratory consultants with extensive experience in the field.





Surgery
Department Oncology

3

a.) Breast Clinic Guide for Patients

The majority of breast lumps are not cancer

Studies:

- 1 in 8 women are affected by breast cancer.
- 23% of women in Kuwait are diagnosed with breast cancer.
- 1% is the risk for men to get breast cancer.
- Breast cancer is number 1 of all cancers affected by women.
- Early detection of breast cancer limits the number of deaths.
- Over 95% survived rate for stage 1 disease.
- Royale Hayat Hospital is the first private hospital to have breast clinic for all its facilities.
- 95% of cases are unknown cause (sporadic) and 5% are related to genetic mutation (BRCA 1 or BRCA 2).
- Breast cancer is the second most common cause of death in women, due to cancer.

Breast clinic at Royale Hayat Hospital

The vast majority of people who are seen at a breast clinic will not have breast cancer. However, it's still important to attend your breast clinic appointment so you can be fully assessed.

What will happen at the breast clinic?

Being referred to a breast clinic might make you feel anxious, so you may want to take a partner, friend or a relative with you for company and support. Your breast clinic appointment may take few hours so that all the necessary tests can be carried out. All tests are available at Royale Hayat Hospital. You usually have a breast examination by a specialized doctor, followed by one or more of the following tests:

- Mammogram.
- Ultrasound breast.
- Core biopsy.
- Fine Needle Aspiration cytology (FNAC).
- MRI Breast.

Visiting the Surgical Oncologist / breast specialist clinic:

History is taken by the breast surgeon: this will include question about any family history of breast problems, any medications you're taking (including hormone replacement therapy (HRT) or history of treatment using fertility drugs contraceptive pill (OCP)) or any previous breast surgery (including breast implants or reduction mammoplasty).

This will be followed by a Breast Examination: in which both breasts are checked by the breast surgeon when you are sitting and when you are lying down. As part of the examination, it's normal to examine the lymph nodes under your arm (axilla) and around the neck. Your assessment may be done in a one-stop clinic. This is where the triple assessment (breast examination, mammogram and tissue biopsy) is arranged during your visit to the clinic with the results available within 3 working days. MRI Breast is also available at Royale Hayat Hospital and the decision to order this test would be done by the Oncologist if further testing of the breast is required.

b.) Breast Self-Examination

Many women find the breast cancer themselves and although BSE is not proven to reduce mortality, women who choose to do it should be informed how to do it effectively. Five simple steps to teach you how to do it:

Step 1:

Begin by looking at your breasts in the mirror with your shoulders straight and your arms on your hips.

Here's what you should look for:

- Breasts that are their usual size, shape and color.
- Breasts that are evenly shaped without visible distortion or swelling. If you see any of the following changes, bring them to your doctor's attention:
- Dimpling, puckering, or bulging of the skin.
- A nipple that has changed position or an inverted nipple (pushed inward instead of sticking out).
- Redness, soreness, rash or swelling.

Step 2:

Now, raise your arms and look for the same changes.

Step 3:

While you're at the mirror, look for any signs of fluid coming out of one or both nipples (this could be a watery, milky, or yellow fluid or blood).



Step 4:

Next, feel your breasts while lying down, using your right hand to feel your left breast and then your left hand to feel your right breast. Use a firm, smooth touch with the first few finger pads of your hand, keeping the fingers flat and together. Use a circular motion. Cover the entire breast from top to bottom, side to side – from your collarbone to the top of your abdomen, and from your armpit to your cleavage. Follow a pattern to be sure that you cover the whole breast. You can begin at the nipple, moving in larger and larger circles until you reach the outer edge of the breast. You can also move your fingers up and down vertically. This up – and – down approach seems to work best for most women.

Be sure to feel all the tissue from the front to the back of your breasts: for the skin and tissue just beneath, use light pressure; use medium pressure for tissue in the middle of your breasts; use firm pressure for the deep tissue in the back. When you've reached the deep tissue, you should be able to feel down to your ribcage.

Step 5:

Finally, feel your breasts while you are standing or sitting. Many women find that the easiest way to feel their breasts is when their skin is wet and slippery, so they like to do this step in the shower.

Cover your entire breast, using the same hand movements described in step 4.

c.) Screening for Breast Cancer

Death rates from breast cancer in Kuwait are considered one of the highest rates in cancers & that is partly due to late diagnosis and lack of awareness.

Royale Hayat Hospital is working on providing patients with general evaluation, rapid and accurate diagnosis of breast problems. Patients can have abnormal findings on a mammographic screening, an ultrasound or even on a clinical examination performed by specialized doctor in breast cancer.

Signs that you should look for and seek medical advice when you have any of the following:

- Breast Lump
- Nipple discharge (bloody or serous).
- Breast changes like skin tethering or nipple inversion.
- Skin changes on the breast like redness, warmth and swelling or edema of the breast.
- Nipple ulceration or scaling.

What are the steps followed for early detection of breast tumors?

1. Clinical examination by specialized doctor in oncology.
2. Mammography starting from the age of 40 every 1-2 years regularly and that is to detect malignant tumors in an early stage as cure rates can reach up to 95% if cancer was diagnosed early, and annually for patients above 50.
3. Ultrasound breast to detect benign breast tumors for patients below the age of 40.
4. Magnetic Resonance Imaging (MRI) of the breasts, mostly used for equivocal cases or those patients younger than 40 years of age and have extensive family history of breast cancer. Usually, the treating doctor is the one who decides when to do this test on regular basis as screening for breast cancer.

How to diagnose breast Cancer?

The best and proper way to diagnose a breast lump is by doing a core biopsy using a special device under local Anesthetic with ultrasound or mammographic guidance. This is done by specialized breast radiologist. The specimen will be examined thoroughly in the pathology department (laboratories) and a diagnosis will be made by the pathologist who will read the slides. This process of preparing and reading the core biopsy takes about 3-5 days. Core biopsies are essential for the treating surgical oncologist in order to decide on the complete and final patient's treatment plan.

d.) Surgical Options in the Treatment of Breast Cancer

Surgical options in the treatment of breast cancer

Today's operations are much less extensive than in the past. When it comes to surgery, "more" is not necessarily "better". Treatment of breast cancer is usually a combination of surgery, radiation, chemotherapy and hormonal therapy.

Why surgery?

Some women may be completely cured by surgery alone. Surgery can offer better control of the cancer. In addition, examination of the tissues removed during surgery provides important information about the type and size of the cancer, the extent of lymph node involvement and the level of estrogen receptors. This information allows the oncologist to determine the stage of the cancer and to tailor any further treatment to your particular case. For some cancers, surgery is not the best treatment. If tumor is large or has very large lymph nodes or it is inflammatory type, chemotherapy should be given first to reduce the size of the tumor. Afterwards, mastectomy may be recommended to further reduce the risk of cancer recurrence in the breast. In cases of metastatic breast cancer, initial treatment.



TYPES OF BREAST SURGERY

Partial mastectomy: removing some normal breast tissue surrounding the cancer. (Also called wide local excision or segmental mastectomy). Studies have found that a partial mastectomy followed by radiation gives a woman the same chance of survival and control of the cancer as does an operation that removes the whole breast.

Total (simple) mastectomy: removal of the entire breast and nipple without removing any chest wall muscles.

Modified radical mastectomy: total removal of the breast along with the axillary lymph nodes. It is called "modified" because it does not remove the chest wall muscles.

Radical mastectomy: removal of the breast, the pectoralis major and minor chest wall muscles and all lymph nodes from the armpit right up to the collarbone.

Skin sparing mastectomy: the breast tissue is "dug out" through a small incision with removal of the nipple-areola complex while preserving the skin.

Staging of the axilla in breast cancer surgery

If the axillary lymph node is proven to be positive by fine needle aspiration cytology, then patient needs to have axillary dissection which is removal of the lymph nodes in the armpit through a separate incision in partial mastectomy or same incision in the case of mastectomy.

Sentinel lymph node biopsy

New technique developed to eliminate the need to remove lymph nodes that do not contain cancer. A radioactive tracer is injected at nuclear medicine department around the areola (periareolar) and /or blue dye. The dye and tracer are absorbed into lymphatic channels and travels to the first lymph node affected by cancer cells from the tumor in the breast. One or two nodes removed and examined intra-operatively by a pathologist using "frozen section" technique. False negative rate should be <10%. This technique helps to reduce the complications found post axillary lymph node dissection (most important is lymphoedema).

e.) Post Breast & Axillary Surgeries

1. Hair-brushing exercise (for hospital)

Rest elbow on bed-table. Keep head erect. Start by brushing one side only, then gradually increase to whole head. Don't overdo, but be persistent.

2. Squeezing and relaxing hand (for hospital)

A rubber ball or similar object may be used.

3. Arm-swinging

Place unaffected arm on back of chair and rest forehead on arm. Allow your other arm to hang loosely and swing from shoulder, forwards and backwards, then side to side and in small circles. As arm relaxes, increase length of swings and size of circles. Swing until arm is relaxed.

4. Bra-fastening

Extend arms, drop hands from elbows, and then slowly reach behind back to bra level.

5. Wall-reaching

Feet apart for balance. Stand close to and facing wall. Start with hands at shoulder level and gradually work hands up the wall. Slide hands back to shoulder level before starting exercise again. Do slowly several times a day. Mark spot reached and aim higher each time.

6. Bean-bag exercise

(a small purse or cosmetic bag will do just as well) Drop bag from right hand over right shoulder into left hand at back. Repeat five times and do with opposite side.

7. Rope-pulley exercise

Throw rope or dressing gown cord over top open door with door between legs. Hold lower end in hand on the side of your surgery and gently pull another end. Raise arm as high as possible each time, until full elevation.

8. Rope/string exercise

Attach rope to doorknob or handle. Make small circles with rope moving entire arm from the shoulder. Do five times on one direction and five times in the other and gradually increase size of circle (by moving in closer) and number of circles.

9. Bean-bag exercise

With towel or similar item use a gentle back-drying motion. Reverse procedure.



f.) Breast Biopsies

What is a biopsy and when is it needed?

If the diagnosis is uncertain after doing the mammogram and ultrasound of the breast, then next step is to take some breast tissue from the area and look at it under a microscope. The process of removing the tissue is called "taking a biopsy".

A fine needle aspiration:

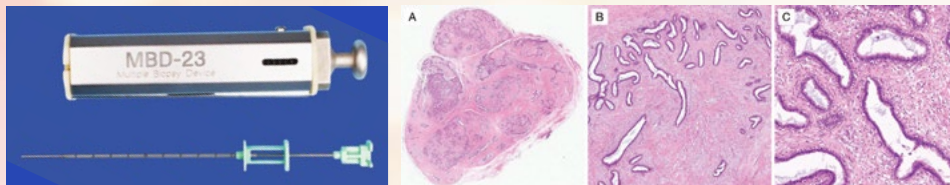
A syringe with very thin needle is used to draw out some material (cells) from a lump the procedure is performed by a radiologist and a cytologist. This can be done accurately under ultrasound guidance usually done for a lymph node.

A core biopsy:

A needle is used to obtain multiple small cores of tissue about 2mm in diameter and is 1-3 cm in length. The needle is slightly wider than a fine needle. The core biopsy needle works by using a spring to advance the needle into an abnormal area to cut small slivers of tissue, anesthetic (freezing) is injected, a small "nick" is made in the skin and the needle is inserted into the breast. When a sample is obtained, the spring makes a loud "snapping" sound after taking the biopsy the radiologist may insert a very small marker inside the breast to mark the location of the biopsy (clip). The marker makes it easier for the surgeon to locate the abnormal area. Core biopsies are usually performed using ultrasound or mammographic guidance.

Once the biopsy is complete the pathologist will examine the tissue under the microscope. The pathology report takes 2-3 days to complete. The use of mammographic guidance requires a special machine called a stereotactic unit attached to the mammogram machine and a special device to do the core needle biopsy. Some stereotactic units are designed for the patient to lay on her tummy with her breasts hanging through a hole in the table. The procedure usually takes one hour.

Breast biopsies are essential for accurate and proper planning of patient surgical and final treatments.



g.) Chemotherapy

A medication to treat disease.

The drugs travel through the blood stream, reaching cancer cells that may be in distant organs. They are called systemic treatments because they attack the cancer through the blood system. In contrast, surgery and radiation are local or regional treatments because the target is one part of the body.

Chemotherapy is used in three ways to treat breast cancer:

1. As adjuvant treatment in early cancer to prevent recurrence (after surgery)
2. As the neoadjuvant or primary treatment for locally advanced (high-risk) cancers to make the size of cancer smaller.
3. To relieve symptoms and possibly improve survival from cancer that has spread to other parts of the body (metastatic cancer).

How long does chemotherapy take?

Usually prescribed for 6 months period. The drugs are given intravenously on one day followed by 21-day drug-free "rest period" and then repeated (3 weekly).

"But I've heard "horror stories" about chemotherapy"

Chemotherapy has a bad name due to the severe side effects people used to experience when these drugs were first being developed. A number of new drugs are used now to eliminate many of these side effects. Once you understand the reason behind the oncologist's choice of chemotherapy, and you know what to expect, it is easier to accept some side effect knowing that the treatment is the best possible choice for your particular situation.

Side effects of chemotherapy

All drugs have potential side effects. What counts is that the beneficial effects of a drug outweigh the problems or discomfort of its side effects. It is important to be aware of possible side effects before beginning treatment and to discuss them with your doctor. This will make the chemotherapy process less mysterious and frightening. Different chemotherapy drugs work in different ways. To take maximum advantage of this, the drugs are often given in combination attacking on all fronts.

Hair Loss

Losing your hair is often the most difficult part of chemotherapy. Hair loss occurs because the chemotherapy slows down the rapidly dividing cells of the roots of the hair. Thinning usually begins about two weeks after the first dose of chemotherapy. You will notice that you are shedding in the shower, on your brush and on your pillow. The hair ALWAYS grows back after finishing treatment.



Hair loss is the most upsetting event for both women and men as it is a public symbol of your cancer. In most situations it cannot be avoided, but fortunately it is temporary. Buy a wig before it happens and take it to your hairdresser to get it styled so you are prepared. If you have long hair, a couple of gradual haircuts could make the change less startling for yourself and your family. Hats and scarves can be very helpful.

Infection

White blood cells in the blood stream protect the body from infection. Many chemotherapy drugs reduce the white blood cell count. If it drops too much, your body's defense mechanisms are low and you have a higher rate of getting an infection.

Your precautions should be:

Avoid crowds and contagious diseases, wash hands frequently, practice good skin care with showers or baths, use of soft tooth brush, use of electric shaver rather than a razor. If you get a fever, sweats, chills, a cough, burning urine, you should call your doctor immediately so antibiotics can be prescribed. The white cell count usually recovers about 21 days after chemotherapy. A hormone called "granulocyte colony stimulating factor" (Neupogyn, G-CSF) may be prescribed if you have infection and low white blood cell count. It helps your bone marrow recover and increases your white blood cell count.

Anemia

Usually, the red blood cells do not drop too much. Anemia may make you feel tired, dizzy, short of breath or chilly. If severe, your doctor will recommend a blood transfusion or injection of "erythropoietin" (Epo, Epogen) which may stimulate your bone marrow to make more red blood cells. Iron supplements are of no benefit.

ABNORMAL BLEEDING AND BRUISING

Nausea and vomiting

May start 6-8 hours after chemotherapy injection or the next day. Antiemetic drugs are prescribed before and after chemotherapy.

Diarrhea and constipation

Drink a lot of clear fluids; avoid foods like cabbage, beans, spicy food. Milk products may contribute to diarrhea.

Sore mouth (mucositis)

Can appear 5 days after treatment. Avoid irritating food: acidic, spicy or rough food. Rinse mouth with baking soda and water. Ask your doctor for advice.

Menstrual periods and sexuality

Chemotherapy may cause your periods to be irregular, to stop permanently during chemotherapy and then return, or to stop permanently. Chemotherapy drugs generally do not affect the ability to have sex, **few changes may be noticed:**

Vaginal dryness, candida infection. Your libido may be affected by stress of the illness, fatigue, self-confidence and your body image.

It is important not to get pregnant during Chemotherapy. These drugs may cause deformity of fetus in the first three months. Oral contraceptives are generally NOT recommended if you have breast cancer because estrogens are believed to stimulate some cancers.

You can discuss with your oncologist with regard to best time to get pregnant after finishing your treatments. If you are pregnant when you are diagnosed with breast cancer, chemotherapy may be given if you are in the second or third trimester.

General symptoms

- Skin changes e.g., dryness, spots, and rash
- Dry, gritty eyes
- Flu-like feeling
- If using Adriamycin which is red, your urine may change color
- Joint or muscle aches and pains
- Tingling in the fingers and toes
- Loss in short term memory (temporary)
- While on chemotherapy, you can eat whatever you like.
- Your chemotherapy drugs are not dangerous to anyone else.



h.) Radiation Therapy

How does radiation work?

"Radiotherapy" is the use of high-energy rays to kill cancer cells. Radiation works by damaging the cells so that they eventually die. Radiation will damage any type of cells, normal or cancerous that lies in the path of the beam, fortunately, normal cells repair themselves from radiation damage, so by giving the radiation in a series of small treatments, usually once a day for a month in case of breast cancer, normal cells have a chance to recover between treatment while cancer cells die.

The procedure for getting radiotherapy

- The treatment decision- usually decided in the breast meeting with consultants in radiation oncology.
- The planning session- this is to plan exactly how the x-ray machine is to be directed. The area to be treated can be marked out using a machine "simulator". Very specific information is recorded during the planning session so that the treatment machine can be accurately positioned each day. CT simulator is used nowadays. The area to be treated is tattooed as reference points for daily treatments. The amount of radiation you receive and number of daily visits depends on the amount of tissue to be treated.

What happens during radiation treatment?

Radiation treatments are painless. You will be in the treatment room for about 10 minutes. During a typical radiation treatment session, the machine is turned on for only 3 minutes per treated area each day and it is given daily. (Except Weekends)

Can radiation be repeated at some time in the future?

No, if the patient received maximum dose for breast radiation, treatment of that area must not be repeated.

Who benefits from radiation therapy?

- After breast-saving partial mastectomy
- After a modified radical mastectomy when there is a high chance of recurrence on chest wall or in the lymph node.
- As treatment for locally advanced cancer after responding to chemotherapy and finishing surgery.
- As the relief of symptoms due to cancer recurrence on the chest wall or metastases to other areas such as the bones, the lymph nodes or the brain.

Side effects of Radiotherapy

Radiation treatments are usually painless. You will not lose your hair. You are NOT radioactive and you are not a threat to your family or children.

Radiation does not make you dizzy. You should feel well enough to drive yourself back and forth to the clinic for your treatments. Some women even continue fulltime employment during radiation treatment.

Side effects include:

- You may feel tired.
- You may feel weepy, depressed, angry or frustrated.

Side effects in the treated breast:

- The skin gradually becomes pink, red or sun burnt and sometimes eventually peels. This usually disappears slowly. Towards end of the course, the skin becomes dry. Ask your doctor what creams you can apply on it, like moisturizing creams, hydrocortisone cream or antibiotic cream.
- Breast firmness-radiation can cause the breast to become enlarged, tender or heavy with fluid during treatment. This may last for 6- 18 months afterwards.
- "Electric shock", prominent blood vessels and scarring. This is NOT a sign of cancer.

Side effects from organs treated within the field:

- Lung (radiation pneumonitis).
- Heart (for the left sided breast cancers).
- Scarring under the arm.
- Scarring on the chest wall.



i.) Hormonal Therapy

What is the relationship between hormones and breast cancer?

Estrogen not only stimulates the growth of the breasts but also appears to encourage the growth of some breast tumors.

Do hormone treatments work for all breast cancers?

Hormone treatment is only effective in tumors that have hormone receptors.

Hormone receptors can be either estrogen receptors (ER) or progesterone receptors (PR) and are found on approximately 70-75% of breast cancers. They are measured on the tumor by the pathologist and predict if the tumor will respond to hormone treatment. In general, the higher the estrogen receptor level in the tumor, the more responsive the tumor will be to hormone treatments. Tumors may have both receptors positive (ER+ PR+), just estrogen receptors positive (ER+ PR-) or both negative (ER- PR-).

What is hormone therapy?

Hormone therapy for breast cancer (also called "anti-estrogen" therapy or aromatase inhibitors) is a form of whole body (systemic) treatment that is used as adjuvant treatment or for cancer recurrence. These include the use of drugs for both premenopausal and post-menopausal women or the removal or destruction of the ovaries in pre-menopausal women by either surgery, medications or radiation.

Hormone therapy is used for prevention, in situ disease, the adjuvant therapy of invasive disease and in recurrent breast cancer. Hormone therapy has been shown to decrease the chance of the cancer coming back both in the affected breast and in the rest of the body, and has been shown to decrease the risk of a new cancer developing in the opposite breast.

Hormone therapy in premenopausal women:

Before menopause, estrogen is made primarily by the ovaries. Regular menstrual cycles are a sign of normal ovarian function. The goal of hormone therapy in the breast cancer is to either stop the production of estrogen from the ovaries or stop the effect of the estrogen on the breast cancer cells.

Tamoxifen

(Novaldex, Tamofen, etc) is effective in stopping the effect of estrogen on the breast cancer cells.

Ovarian Suppression

Various treatments may put a woman into menopause either permanently or temporarily:

1. Oophorectomy – surgical removal of the ovaries which can be done with minimal surgery (laparoscopically).
2. Radiation ablation- using radiation (usually daily for five days) to permanently damage the ovaries so they stop producing estrogens. It generally takes about 3 months after radiation to the ovaries to decrease levels of estrogen to a post-menopausal level.
3. Pituitary suppression – drugs called (LHRH agonist) given as intramuscular injection every month to three months for 2-5 years. These drugs include goserelin (Zoladex) or buserelin (superfact).

Hormone therapy in post-menopausal women:

A woman is defined as post-menopausal when she has not had periods for a year. Menopause can be confirmed by blood test for hormone levels. After menopause a woman still makes estrogen but the levels of estrogen are significantly lower than before menopause.

In post-menopausal women, hormone sensitive breast cancers can be treated either by blocking the estrogen receptor on the cell (tamoxifen) or by interfering with the aromatase enzyme to stop estrogen production with a group of drugs known as aromatase inhibitors: anastrozole (arimidex), letrozole (famara) and exemestane (aromasin).

Side effects of Hormonal Therapy

Some women have no side effects while others suffer from it. This usually cannot be predicted.

Menopausal symptoms:

- Hot flushes – most women report they get better with time.
- Vaginal dryness and irritation – can make sexual intercourse painful and difficult. Lubricating jellies may be helpful.
- Osteoporosis and increased risk of fractures: all premenopausal women should take in 1000mg of calcium and 400 IU of vitamin D per day and postmenopausal women take in 1500mg of calcium and 800 IU of vitamin D per day. For women on aromatase inhibitors, a baseline bone density should be done and repeated every 18-24month.
- Weight gain.



Tamoxifen

Anti-estrogen, dose of 20mg per day for five years, side effects include:

- Hot flashes
- Vaginal discharge: clear or whitish
- Phlebitis and /or thrombosis: risk of deep vein thrombosis or pulmonary embolism (1% risk).
- Continuation of menses: birth control precautions should be maintained as tamoxifen could be harmful to a fetus.
- Endometrial (uterus) cancer: two women per 1000 each year
- Very rare side effects include: nausea, vomiting, loss of appetite, muscle and joint pains, rashes and skin dryness, headache and dizziness, facial hair and vision changes.

Aromatase Inhibitors

Only effective in post-menopausal women, e.g., Arimidex, femara. The most side effects are those associated with menopause including hot flushes, vaginal dryness and increased risk of osteoporosis.

- Muscle and joint aches and pains – anti-inflammatory pill such as ibuprofen might be helpful.
- Increase in lipid and triglyceride levels
- Loss of sexual libido.
- Less common side effects include: nausea, headaches, swelling of hands and feet, fatigue and thinning of the hair, skin rash, depression, weight gain.





Diagnostic Imaging

4



Breast imaging available at Royale Hayat Hospital

Breast imaging is the examination of the breast to demonstrate any changes and abnormalities using mammography and /or ultrasound.

A mammogram is an x-ray test of the breast tissue. It can provide information about lumps, calcification and other abnormalities that may be present in the breast. It is also available for screening purposes for women without any breast problems over the age of 40.

A mammogram is a safe procedure. It is essential that you tell the Radiographer if:

- You have breast implants.
- You are pregnant.
- You think you might be pregnant.

Each breast needs to be compressed between the x-ray machine and a flat x-ray plate. This pressure is important to get good results. It is NOT painful but it can be uncomfortable.

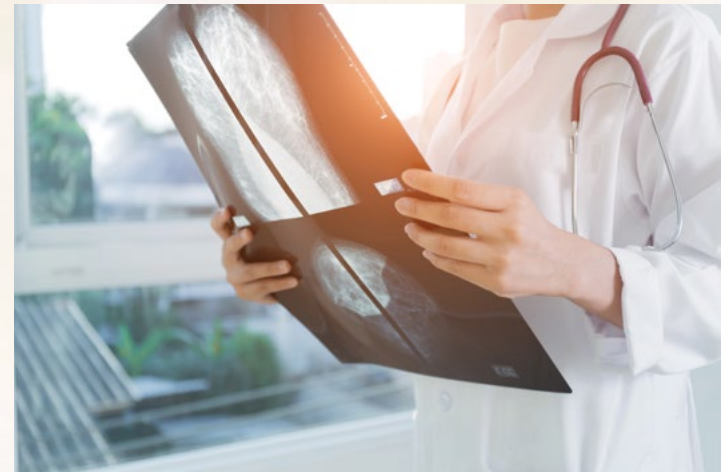
Two x-rays' views are taken of each breast from different angles. Each mammogram exposure takes only few seconds. You will be in x-ray room for approximately 20 minutes.

Breast ultrasound:

Uses sound waves – no radiation – to show breast tissue. It can provide us with further information. A gel will be applied to your breast and a small hand-held sensor will be moved over the skin to view the breast from different angles.

Core biopsy and / or Fine Needle Aspiration: if you have a solid lump, the doctor will need to do a core biopsy which is a needle that is inserted into the breast lump to get a sample (a piece of tissue) of it. Local anesthetic is used prior to insertion of the needle to numb the area.

On the other hand, fine needle aspiration (FNA) is used when few cells from lymph node in the axilla are removed through a smaller needle. The tissue or cells are sent to the laboratory and examined under the microscope. It takes about two to three working days for the results to be ready.





Nutrition Care

5

What is the rate of breast cancer prevention?

There is no prevention for breast cancer as such by adapting a healthy lifestyle can lower your risk of having breast cancer.

What is the incidence rate of breast cancer?

Breast cancer is the most common cancer among women around the world. Around 1 in 8 women may develop the disease.

Breast Cancer Risks

Certain risk factors can increase a woman's chance of getting breast cancer. Having one or more of these risks doesn't mean that you'll get breast cancer. Sometimes a woman gets breast cancer without any of the risk factors (other than being a woman and growing older).

Age

Is the strongest risk factor for breast cancer, and aging increases the risk.

Sedentary lifestyle

A sedentary lifestyle increases the risk of developing breast cancer.

Family history

Inheriting the gene increases cancer risk by (5-10%).

Body weight

Being overweight or obese after menopause, increases the risk of breast cancer.

Alcohol

Excess intake of alcohol increases the risk of breast cancer.

Hormone Replacement Therapy and fertility drug treatments

Increase the risk of breast cancer.

Nutrition is part of breast cancer prevention

Breast cancer prevention through adapting healthy lifestyle

Eat a healthy diet

Eating a healthy, balanced diet may lower the risk of developing breast cancer by 10-20%.

Maintain a healthy weight

Excess fat tissues (obesity) may increase the chances of developing breast cancer by raising the levels of estrogen, insulin and other hormones.

Breastfeeding

The longer you breastfeed, the greater the protective effect.

Break the habit

The risk of developing breast cancer in women is 24% higher in smokers than in non-smokers.

Exercise regularly

Physical activity helps maintain a healthy weight which, in turn, helps prevent breast cancer. Exercise 30 minutes daily reduces breast cancer risk by 18%.



Breast cancer prevention through adapting a healthy diet

Avoid sodium in cooked and preserved food

Eat foods low in sodium and choose products with no more than 140 mg per serving by reading the food label.

Limit consumption of red meats and avoid the processed.

Eat no more than 500 grams cooked weight of red meat per week, like beef and lamb, and avoid processed meats such as bacon, salami, hot dogs and sausages.

Limit foods and drinks high in fat or sugars

Limit caloric dense foods such as fast foods, fries, and dressings. Cut down on sugary drinks such as carbonated and energy drinks, and canned juices.

Substitute animal-based protein with plant-based protein

Examples of plant-based proteins: lentils, black and green beans, quinoa, chickpeas, green peas, almond, artichoke, spinach, broccoli and asparagus.

Increase antioxidant intake from natural sources

Increase your fruit and vegetable intake and avoid the use of supplement.

Fiber Up

Increasing intake of daily fiber can decrease breast cancer by 7%.

Get your vitamin D

Maintaining adequate levels of vitamin D is associated with lower risk of developing breast cancer. High sources of vitamin D include: salmon, tuna, fortified milk, and fortified cereal.

MYTHS AND FACT

Does flaxseeds increase the risk of breast cancer recurrence?

- No studies have shown that flaxseeds may increase the recurrence of breast cancer.
- It is safe to be consumed in moderation.
- 1 to 3 tablespoons daily may protect against breast cancer.
- Flaxseed is an excellent source of magnesium, manganese, thiamin, copper, protein, omega-3, and fibers.



Are soy foods safe for breast cancer survivors?

- Breast cancer survivors can safely eat soy foods in moderation.
- Moderate consumption is 1 to 2 servings/day of soy foods, such as tofu, soy milk, edamame and soy nuts).
- 1 serving of soy equal 1 cup of soy milk - ½ cup of cooked soy beans - 1/3 cup or 28 grams of soy nuts.
- Excessive intake of Soy foods may increase the risk of recurrence in breast cancer.



Does sugar feed cancer cells?

- There is no direct relation between sugar intake and the cancer cells.
- It is recommended to limit sugary foods as part of a healthy diet to avoid putting on weight.
- High sugar intake may contribute to excess weight gain and obesity, which is associated with an increased risk of developing breast cancer.



Does dairy products intake increase the risk of breast cancer?

- There is not enough evidence that link dairy product intake to breast cancer.
- Dairy product is a good source of calcium and protein, which is important for building bones and teeth and help to repair muscles.
- Choosing low fat and low sugar dairy products help to keep a healthy weight, which reduce the risk of cancer.





Royale Hayat Dental

6



BREAST CANCER AND ORAL HEALTH

Do you know that there is surprisingly a close connection between oral health and breast cancer?

Gum disease increases the risk of developing breast cancer.

Women with gum disease have a 14 % overall increased risk for breast cancer compared to women without gum disease.

THINGS TO DO TO PREVENT GUM DISEASE

- Brush your teeth with fluoride toothpaste at least twice a day.
- Clean in between your teeth every day using floss or interdental brushes.
- Replace your toothbrush every 1 to 3 months.
- See a dentist and dental hygienist for regular check-ups, especially if you're pregnant or have type 2 diabetes.
- Advise you to stop smoking, if you smoke.

IF YOU DIAGNOSED WITH BREAST CANCER

Take action for your oral health before and during treatment:

Talk to your dentist about your diagnosis as soon as possible. You may need to schedule a dental appointment prior to beginning treatment, especially if you haven't had a checkup recently.

Undergo any dental procedures you may need before treatment starts and the immune system's and body's ability to heal itself is compromised.

Allow time to heal from any dental treatment before beginning cancer treatment, if possible. Your dentist can recommend how much healing time you need between dental procedures and receiving cancer treatment. Connect your dentist and physicians.

Your dentist should be aware of your treatment plan to better serve your needs, and your cancer care team should know how treatment impacts your mouth.

IF you have breast cancer, take a look of following information.

Breast cancer treatments can cause oral side effects such us:

1. Dysgeusia - metallic, salty, foul or rancid taste sensations.
2. Xerostomia - dryness of the mouth or thickened saliva.
3. Stomatitis - mouth sores and ulcers.
4. Oral mucositis - inflammation and ulcers of gums.
5. Dysphagia - difficulty or discomfort swallowing.
6. Dental caries – cavities.
7. Infections of the mouth and / or throat.
8. Trismus- also known as lockjaw.
9. Candida infection - oral thrush or yeast infection.

If you have any of the previous side effects, please contact your dentist for treatment to relieve your symptoms.





La Cosmetique Royale

7

Breast Reconstruction Surgery at La cosmetique Royale (Cosmetic Center)

Women who have surgery as part of their breast cancer treatment may choose breast reconstruction surgery to rebuild the shape and look of the breast. There are different types of breast reconstruction. Learn about all your options and what to expect before and after your surgery.

Breast Reconstruction Options

Women who have had surgery to treat breast cancer can choose from several types of breast reconstruction. When deciding what type is best for you, you and your doctors should discuss factors including your health and your personal preferences. Take the time to learn what options are available to you and consider talking to others who have had that procedure before you make a decision.

Types of breast reconstruction procedures

Several types of reconstructive surgery are available, and often the process means more than one operation. Give yourself plenty of time to make the best decision for you. You should make your decision about breast reconstruction only after you are fully informed.

The two main types of breast reconstruction are implant reconstruction and tissue (flap) reconstruction. Sometimes the implant and flap procedures are used together to rebuild a breast.

For all types of breast reconstruction there is often an opportunity for future “touch up” procedures, such as fat grafting and scar revisions. Breast reconstruction can also recreate the nipple-areolar area that can be done with a small surgical procedure, tattooing, or a combination of both. This is done to help make the reconstructed breast look more like the original breast.

For more details & inquiry call us at 00965 2536 0555





Royale Home Health
Breast Cancer Program

8



Experiencing breast cancer is a challenging and stressful time for a woman and her family. Ensuring that you have the right psychological and medical support throughout your care journey will help you better to manage your outcomes and possibly recover faster. The Royale Home Health breast cancer program is specifically designed for this purpose. You will receive clinical care that is based on evidence and international practice.

We offer care based on your specific requirements and preferences including Doctor Home Visit, Doctor Video Consultation, Palliative Nurses Visits and Rehabilitation Support. All care will be coordinated by our Dr Ameena Mohammed Al Ansari, Consultant of Family Medicine & Palliative Medicine.

Doctor Stream

Our Palliative Specialist Doctor will visit your home to do a thorough breast assessment, also will teach the client how to do a self-breast examination, provide psychological counselling, prepare a customized treatment plan and a follow-up management including breast ultrasound recommendation.

Nursing Stream

The specialty nursing visits include in - depth assessment of the patient with a care plan, education sessions, and dressing changes and drain care as well as emotional support with the nurse throughout the journey. The nurse will visit you pre and post-surgery in the comfort of your home.

Rehabilitation stream

The physiotherapist will provide customized exercises specific for pre and post - surgical breast cancer including guidance and assistance with lymphatic swelling depending upon the surgery. In addition to that under the supervision of physiotherapist there will be therapeutic massages provided by the experienced massage therapist.

Our multidisciplinary team offer you a customized, holistic care to your needs throughout your journey.



